

Section: **3.1 – MECHANICALLY FASTENED JOINTS**

Chapter: **3214**

Title: **RIVETS LOADS IN ECCENTRIC 3D JOINTS**

Revision: **1.0**

Date: **28/02/2018**

1 INTRODUCTION

This chapter defines the methodology to use for obtaining the loads in the rivets or bolts, which make up the mechanically fastened attachment between a fitting and a support structure, when not all the fasteners are installed in just one plane, and the utilization of other more sophisticated methods (like numerical FEM analysis) are not appropriate.

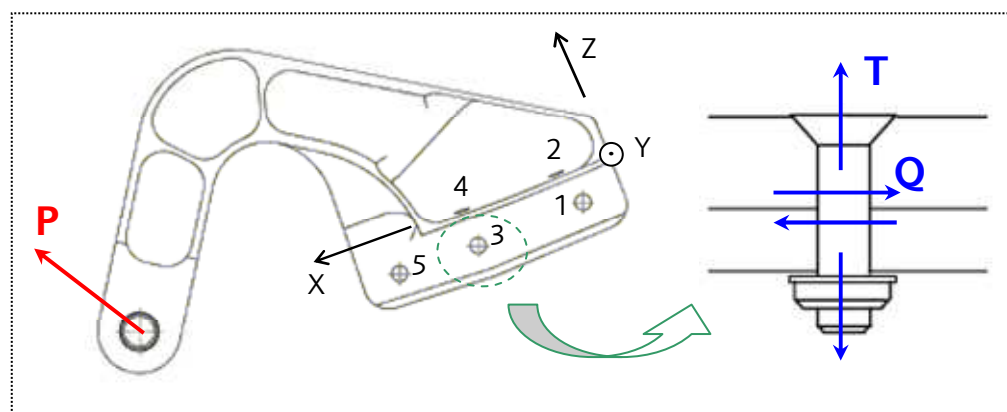


Figure 1. Sketch of fitting and fastened joint to several planes

2 METHOD DESCRIPTION

The non-numerical method proposed herein is supported in two main assumptions: the fitting behaves as a rigid-body, and the point-stiffness at each rivet position is known. The first assumption generally fits well with the behaviour of real well designed parts, i.e. fittings made of machined metallic alloys, with sound robust loads paths from load application points to attachment rivets. The estimation of the

stiffness of each rivet, in particular along the direction of its axis (stiffness in tension) is a more complex topic, dependent on the rivet itself, on the flange of the fitting where it is installed, and on the supporting structure. Expressions for estimating rivets stiffness were presented in previous chapters of this manual (see chapter 3210 "CALCULATION OF RIVETS LOADS").

The method for calculating these loads is based in relationships defined by rigid body kinematics. Those relationships, plus the global equilibrium of forces, allow to calculate the action forces at each of the fastening points of the fitting, if a force is applied in another point of that fitting / rigid-body. That approach is a possible mathematical simplification for the calculation of three-dimensional joints of fittings.

For those familiar with Finite Element Models solved with MSC-NASTRAN, the methodology presented would be equivalent to an analysis with a RB2 element representing the fitting, its independent node at the point of application of the external forces, the dependent nodes at rivets positions, and the rivets nodes attached to ground by means of CELAS1 elements.

Detailed description of the method is presented in document:

NT-T-ADP-09003 Is. C "Theoretical Manual of Program UNIONES. Versions 1-2." (Ref. 1)

This methodology is currently implemented in Airbus DS software program UNIONES (version 2.0).

3 SCOPE AND LIMITATIONS

Calculation of the loads in the rivets attaching a fitting to a structure, when the rivets distribution is not enclosed within just one plane.

- ✓ Fitting assumed stiff as a rigid body
- ✓ Linear elastic relationships between displacement and forces at rivets nodes
- ✓ Contact forces between plates not computed with a dedicated stiffness (in negative tension loads)

The stress engineering should take special care if multiple load paths exist e.g. if the loaded part is joined to more than one part. In this case the individual stiffness/compliance of each part may significantly alter the load path compared to the assumed rigid body solution.

This method is not recommended for final certification analysis of three-dimensional fastened eccentric joints unless the analysis parameters (the rivets stiffness, especially in tension) for the case of study, or for a reasonably similar configuration, were previously validated by precise means (FEM numerical simulation or tests).

One option for the analyst to mitigate the risk on non-conservative inaccuracies when using this method (not feasible for all types of rivets arrangements), is to run two analyses for a load case: one with rivets stiffness estimated according to formulas previously discussed, another with stiffness in tension made null, and then using the most critical forces obtained in both analyses to perform the strength check.

4 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
FEM	-	Finite Element Model
Q	N	Shear load in a rivet
T	N	Tension load in a rivet

REFERENCES

<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1 NT-T-ADP-09003	Is. C	Theoretical Manual of Program UNIONES Versions 1-2

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
UNIONES	2.0	Loads in the rivets attaching a fitting to a structure, when the rivets distribution is not enclosed within just one plane.
NASTRAN (MSC-Software)	2005	Multidisciplinary structural analysis software based in the Finite Elements methods

Table 1. *Software programs referenced in this chapter*

DISTRIBUTION LIST

<i>Department</i>	<i>Name</i>	<i>Full Doc.</i>	<i>Cover Page</i>
Stress Methods & Optimisation (TEASP-TL3)	Dr. Fernaß Daoud	X	
Stress Methods & Optimisation (TEASP-TL3)	Dr. Alexander Hörmann	X	
Stress M&L, Derivatives, Mission (TEASA-TL2)	Miguel Angel Calero Gallardo	X	
Stress Combat MAN (TEASA-TL3)	Sebastian Rönz	X	
Stress Combat, Aerostructures, A400M and Technologies GET (TEASA-TL4)	Sofia Ponce Borrero	X	
Structural Repair GET (TEASA-TL6)	Jose Luis Melchor Martin de la Sierra	X	

Table 2. *Distribution list (of electronic document, ahead of public release)*

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub- chapters affected
1.0 28/02/2018	G. Baños F. Glock	First issue	All pages

Table 3. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Analysis (TEASA)	M. A. Calero 26/02/2018	B. Mayer 28/02/2018
Stress Methods (TEASP-TL3)	G. Baños 19/02/2018	F. Glock 20/02/2018

Table 4. *Approval signatures table for last revision of this chapter*